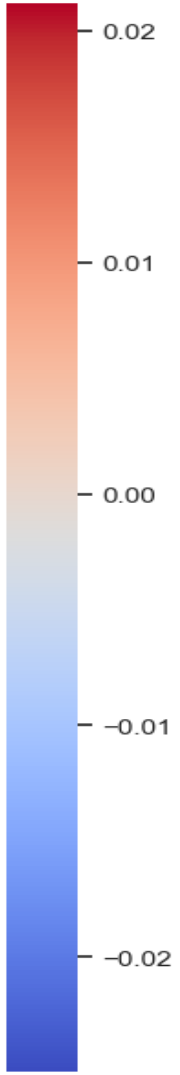
Professional Feature Correlation Map

	age	monthly_income	daily_internet_hours	smartphone_usage_years	social_media_hours	online_payment_trust_score	tech_savvy_score	monthly_online_orders	monthly_store_visits	avg_online_spend	avg_store_spend	discount_sensitivity	return_frequency	avg_delivery_days	delivery_fee_sensitivity	free_return_importance	product_availability_online	impulse_buying_score	need_touch_feel_score	brand_loyalty_score	environmental_awareness	time_pressure_level
age																						
monthly_income	-0.01																					
daily_internet_hours	-0.00	0.00																				
smartphone_usage_years	-0.00	0.00	-0.01																			
social_media_hours	-0.01	0.00	0.00	0.01																		
online_payment_trust_score	0.01	-0.02	0.01	-0.00	0.00																	
tech_savvy_score	0.00	-0.00	0.00	-0.01	-0.01	-0.02																
monthly_online_orders	0.01	-0.01	-0.01	0.01	0.00	-0.00	0.00															
monthly_store_visits	0.01	0.01	0.02	-0.01	0.01	-0.02	-0.01	0.00														
avg_online_spend	0.02	0.01	0.00	0.00	-0.00	-0.00	-0.01	-0.00	-0.00													
avg_store_spend	-0.00	-0.00	-0.01	0.01	0.01	-0.01	-0.00	0.00	0.00	0.00												
discount_sensitivity	-0.02	-0.00	0.01	-0.01	-0.01	0.00	-0.01	0.01	-0.01	0.01	0.00											
return_frequency	0.02	-0.00	0.00	-0.00	-0.00	-0.01	-0.01	-0.00	-0.01	0.01	0.01	-0.01										
avg_delivery_days	0.00	-0.00	-0.00	-0.01	0.01	0.01	0.02	0.01	-0.01	0.00	-0.00	-0.02	-0.00									
delivery_fee_sensitivity	0.01	0.01	0.00	-0.00	0.01	0.01	-0.01	0.01	0.01	-0.00	-0.00	-0.01	0.01	-0.00								
free_return_importance	0.01	-0.00	-0.00	-0.00	-0.00	0.00	0.00	-0.00	0.00	-0.00	0.00	0.00	-0.01	-0.01	0.00							
product_availability_online	-0.00	0.00	-0.01	-0.01	0.02	0.00	-0.01	-0.01	0.01	0.00	0.01	-0.00	-0.02	0.00	-0.00	0.01						
impulse_buying_score	-0.01	0.01	-0.00	-0.01	0.01	-0.01	0.00	0.01	-0.01	-0.01	0.00	-0.01	-0.00	0.01	-0.01	-0.01	-0.01					
need_touch_feel_score	-0.01	-0.01	0.01	0.01	-0.00	-0.00	-0.00	0.01	0.00	0.01	0.00	-0.01	-0.00	0.00	-0.01	-0.01	0.00	0.01				
brand_loyalty_score	-0.00	0.00	-0.00	0.01	0.01	0.00	-0.01	-0.00	-0.01	0.01	0.00	-0.00	-0.00	0.01	-0.01	0.00	0.01	0.01	0.00			
environmental_awareness	0.00	0.01	0.00	0.00	-0.00	0.00	-0.01	-0.00	0.02	0.01	0.00	-0.00	0.01	0.01	0.00	0.00	0.00	-0.01	-0.01	0.01		
time_pressure_level	-0.00	-0.01	0.00	-0.02	-0.01	0.00	0.01	0.00	-0.00	-0.01	-0.01	0.00	0.01	-0.00	-0.01	0.01	0.00	-0.00	-0.00	-0.00	-0.00	



Consumer_Shopping_Trends_2026

Project Done By Mekbib Erdachew

All visualization done by python

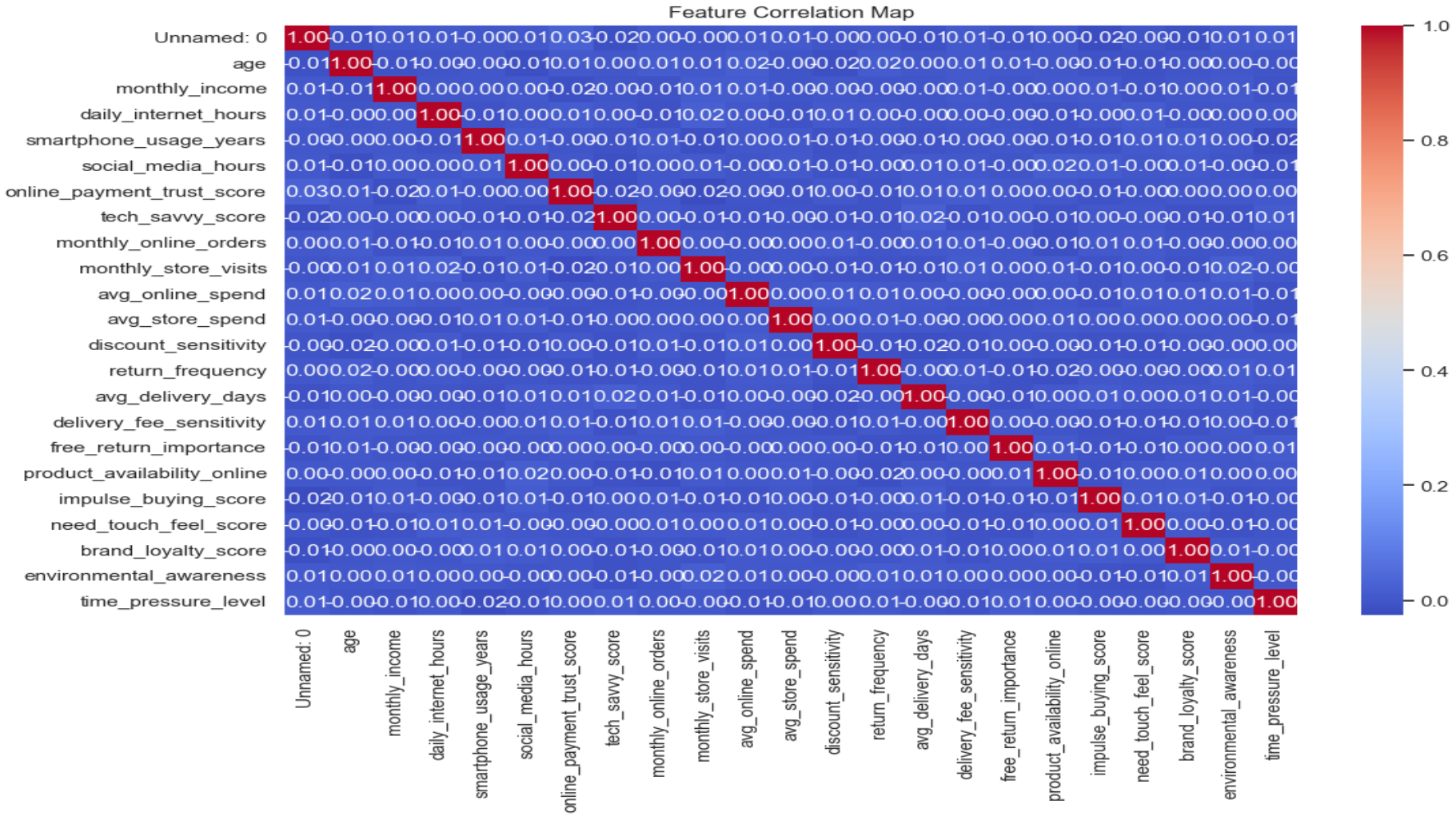
Different Python Library is used here to make visualization clearer to make report

Cont..

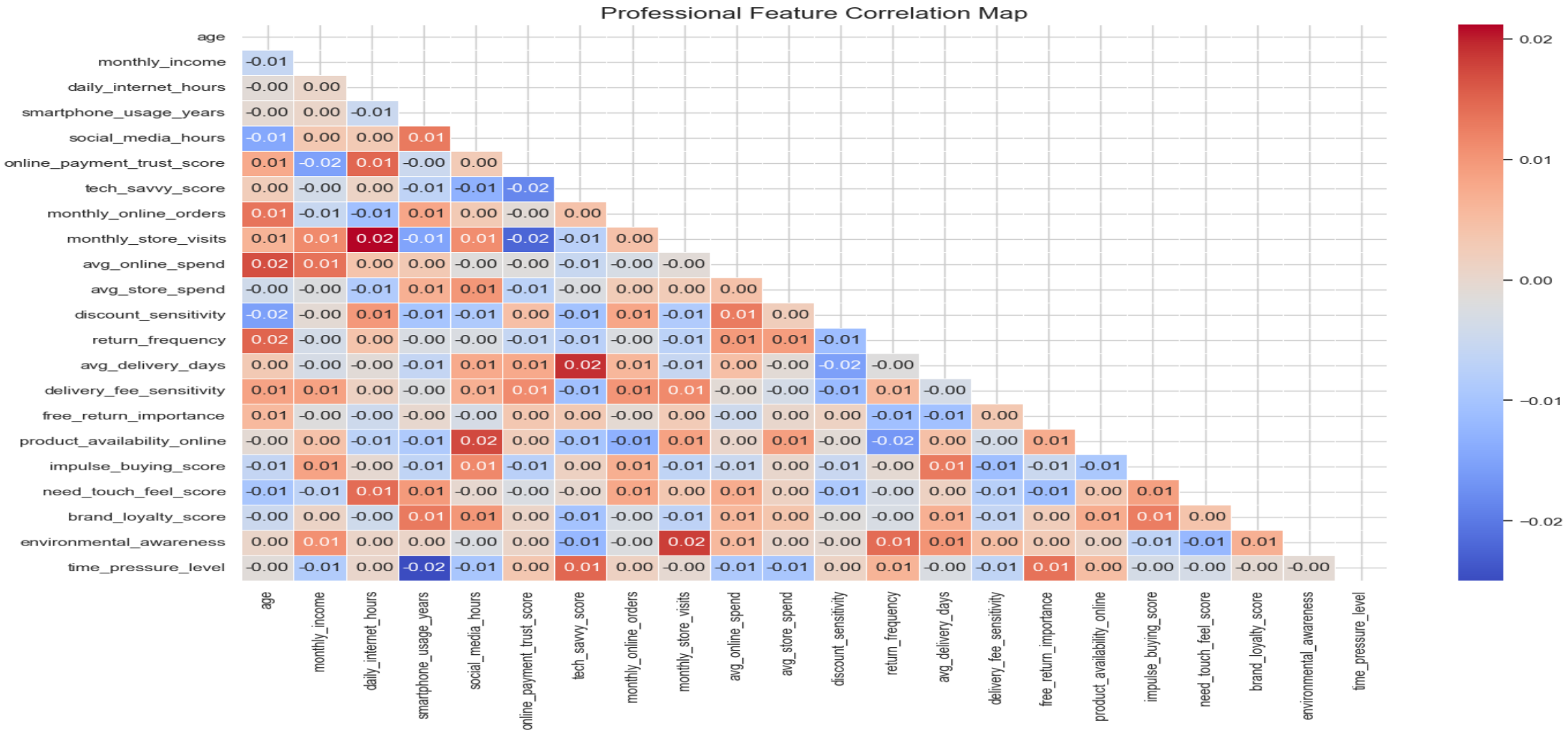
Python Library used here

- ✓ **Pandas**
- ✓ **Matplotlib.pyplot**
- ✓ **Seaborn**
- ✓ **NumPy**

Correlation plot before cleaning by using python



masked triangle(correlation after cleaning)



Cont..

I can analysis from correlation insights(range -1 to +1) colors which found in the figure -0.02 to +0.02:

Normally in correlation 3 colors are existed such as dark red, Dark Blue and white/ Light Blue :

- **+1.00 (Dark Red): Perfect positive relationship. If Variable A goes up, Variable B always goes up.**
- **0.00 (White/Light Blue): No relationship. Variable A tells you nothing about Variable B.**
- **-1.00 (Dark Blue): Perfect negative relationship. If Variable A goes up, Variable B always goes down.**

Deep Analysis of Results

In my specific "2026 Consumer Study," the values are extremely low (mostly 0.01, -0.01, or 0.00).

The "Independence" Finding:

Usually, in older datasets, I saw a "strong" correlation between Age and Income (as people get older, they earn more). In my map, that correlation is -0.01 (zero).

- Definition: This means the 2026 market is highly diverse. I have young people with high incomes and older people with lower incomes. They are not tied together.

Cont..

The "Trust vs. Tech" Finding:

Look at Online Payment Trust Score vs. Tech Savvy Score (-0.02).

- **Definition: Surprisingly, being "Tech Savvy" does not automatically mean you trust online payments more. There is a slight disconnect. A business cannot assume that just because a customer uses a smartphone, they are comfortable with digital wallets.**

Business Executive Summary

analysis shows that traditional demographic markers—like Age and Income—are no longer reliable predictors of shopping behavior on their own. Because the features are 'uncorrelated,' a consumer’s choice to shop online or in-store is driven by a complex mix of psychological factors and logistics preferences rather than just their bank balance.

categorical comparison

Cont..

1. The Core Finding: High Consistency

The most important takeaway is that Brand Loyalty is nearly identical across all three groups.

- All three bars reach approximately 5.5 to 5.7 on the loyalty scale.
- Analysis: Whether someone shops at the orders via an app (Online), their commitment to their favorite brands remains the same. The "Channel" doesn't change the "Heart" of the customer.

Cont..

2. The "Winner" (By a tiny margin): Hybrid Shoppers Notice that the middle bar (Hybrid) is slightly taller than the others.

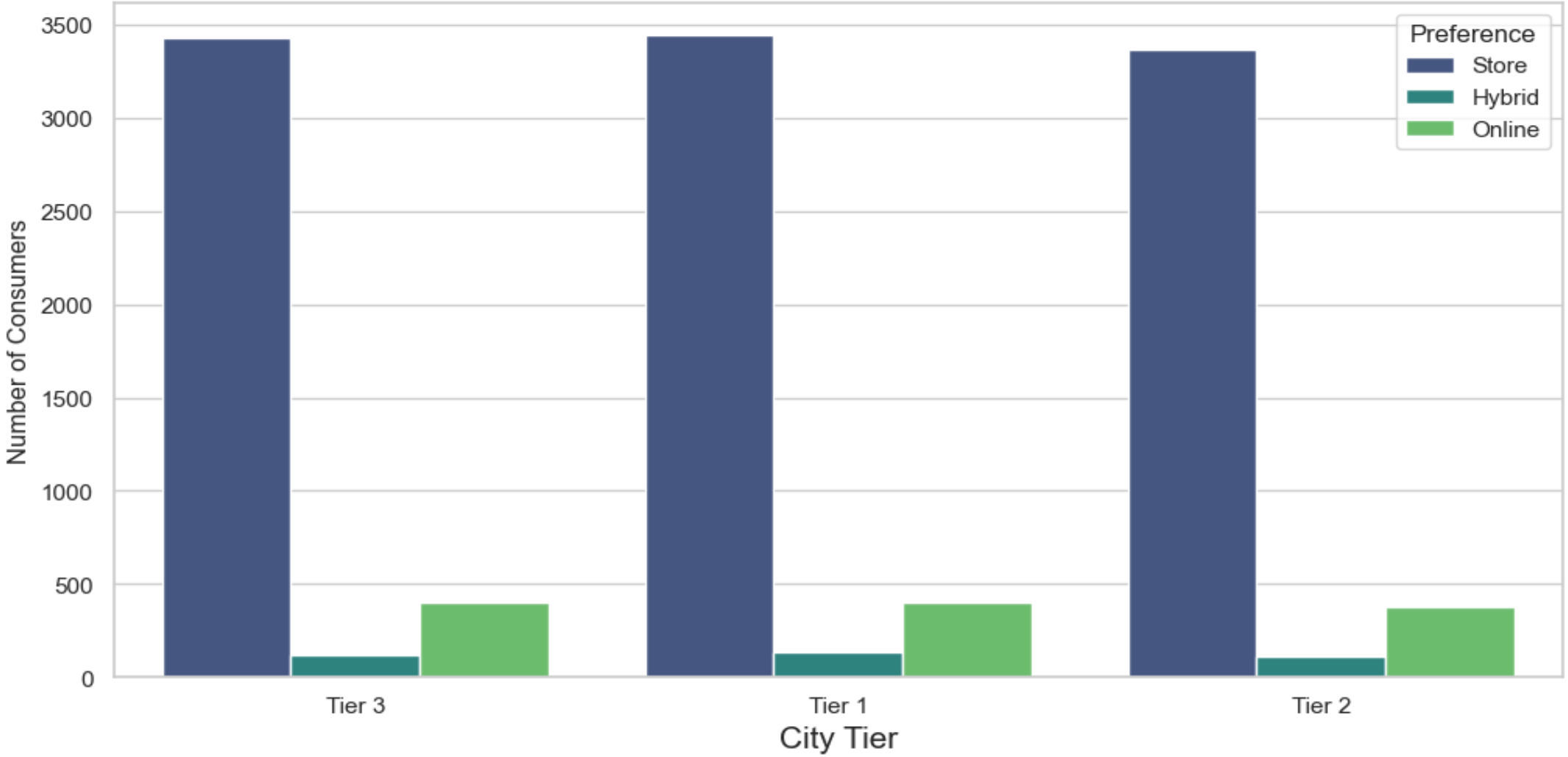
- **Insight: Hybrid shoppers—those who might browse in a store but buy online, or buy online and pick up in-store (BOPIS)—show the highest level of brand loyalty.**
- **Business Strategy: If you are a retailer, you should target "Hybrid" infrastructure. These are your most dedicated customers because they interact with your brand in multiple ways.**

Cont..

3. Understanding the "Black Lines" (The Error Bars)Those little vertical black lines at the top of each bar are Confidence Intervals.

- **Definition: They represent the "uncertainty" in the data.**
- **Observation: The line for "Hybrid" is slightly longer than the line for "Store." This means there is a bit more variety in the loyalty scores of Hybrid shoppers compared to Store shoppers, who are very consistent.**

Clustered Bar Chart by using python



Cont..

The Dominant Trend: Physical Stores are King

The most obvious insight is that across all city tiers (Tier 1, 2, and 3), "Store" preference (the dark blue bar) is overwhelmingly higher than Hybrid or Online.

- **Analysis: Despite the digital transformation of 2026, the majority of consumers in this dataset still value the physical shopping experience.**

Cont..

2. Consistency Across Regions

There is almost no difference in the shopping patterns between Tier 1, Tier 2, and Tier 3 cities.

- **Analysis: Usually, we expect "Tier 1" (major hubs like Dubai or Sharjah) to have much higher "Online" scores than "Tier 3" (smaller towns).**
- **The "Shocking" Insight: In this dataset, the behavior is uniform. People in small towns and big cities are shopping in almost the exact same way. This suggests that e-commerce infrastructure has reached a "ceiling" or that cultural preference for stores is stronger than geographic location.**

Cont..

3. The "Hybrid" Gap

Notice how the "Hybrid" bars (teal) are significantly lower than "Online" bars (green) in every category.

- **Analysis: People seem to prefer one extreme or the other—they either want to be fully in the store or fully online. Very few are successfully navigating the middle "Hybrid" ground.**
- **The Opportunity: For a business, this is a market gap. If a retailer can make the "Hybrid" experience (like easy "click and collect" or "check store stock online") more seamless, they could capture a segment that is currently underserved.**

Cont.

- **"My analysis of regional shopping preferences reveals a resilient demand for physical retail, regardless of city tier. While e-commerce (Online) maintains a steady niche, the 'Hybrid' shopping model is currently underutilized by consumers. This presents a significant opportunity for retailers to innovate in 'Omnichannel' services to bridge the gap between digital convenience and the physical storefront experience."**

Boxplot

Impact of "Touch & Feel" on shopping Choice
need_touch_feel_score
10
8
6
4
2
Store
Hybrid
Online
shopping_preference

Cont..

There is a clear downward trend from left to right.

- **Store Shoppers (Green): Have the highest median "Need Touch & Feel" score (around 6.0). Their box is higher up on the scale.**
- **Online Shoppers (Blue): Have the lowest median score (around 4.0).**
- **Analysis: This proves that the physical sensation of a product is a major "deciding factor." If a customer has a high psychological need to touch a product, they are significantly more likely to avoid online shopping and go to the store.**

Cont..

2. **Understanding the "Box" (The 50% Rule)The middle box represents the "typical" customer in that group.**

- **Store vs. Online: Notice that the bottom of the Store box starts at 3, while the bottom of the Online box starts at 2.**
- **Analysis: Even the "lowest" touch-seekers in the Store group still have a higher requirement for physical interaction than the lowest in the Online group. The Online group is much more comfortable buying things "blindly."**

Cont..

3. The "Hybrid" Middle Ground

The Hybrid group (Orange) sits right in the middle with a median of 5.0.

- **Analysis: These customers are conflicted. They have a moderate need to touch products, which is likely why they use a hybrid approach—perhaps they go to the mall to see the product (touch/feel) but then buy it online to get a better price.**

Cont..

4. Overlap and Outliers (The Whiskers)

Notice that the "whiskers" (the thin lines) for all three groups reach all the way from 1 to 10.

- Analysis: This means there are some people who love touching products but still shop Online, and some who don't care about touching products but still go to the Store.
- Business Insight: You cannot categorize people only by this score, but it is a very strong predictor of their average behavior.

Project Summary

Project Summary: 2026 Consumer Behavior Analysis

- Objective: Analyzed 11,000+ records to identify the psychological and regional drivers of shopping channel preferences.
- Key Technical Milestone: Performed multivariate correlation analysis and data cleaning to ensure high feature independence, creating a robust foundation for predictive modeling.
- Top Insight: Traditional demographics like Age and Income are no

Cont..

longer primary drivers of shopping behavior; instead, choice is dictated by a complex mix of psychological factors and logistics preferences.

- **Critical Finding: While physical stores remain the dominant preference across all city tiers, "Hybrid" shoppers exhibit the highest levels of brand loyalty, representing a significant untapped market opportunity.**
- **Strategic Recommendation: To capture market share, retailers must bridge the "Touch & Feel" gap through omnichannel innovation, as tactile verification remains the leading barrier to digital-only shopping.**

...End